SPI Peripheral RTL Linting & Verification

SONIC-MCU

Static Analysis and Superlint | VERIF-LINT-v1.0

Introduction

This document details the RTL static analysis and linting results for the peripheral IPs of the SONIC-MCU. To ensure robust design practices and synthesizeable code, the Cadence JasperGold Superlint App is utilized.

Currently, this report covers the **SPI Master** IP. Future revisions of this document will expand to include other standard peripherals, such as I2C, UART, and Timers.

Superlint Methodology & Setup

All peripheral modules are evaluated using a unified JasperGold TCL script. The script restricts the verification domain exclusively to LINT checks, disabling other domains to isolate structural and coding style violations.

JasperGold TCL Script (`lint.tcl`)

The configuration script initializes the environment by clearing prior states, loading the file list via `jg_file.f`, and elaborating the top module. The clocks and resets are declared, and a detailed violation report is exported to a text file.

Listing 1: Superlint Extraction Script (`lint.tcl`)

```
clear -all
config_rtlds -rule -disable -domain {all}
config_rtlds -rule -enable -domain {LINT}

analyze -sv -f jg_file.f
elaborate -top apb_spi_master

clock HCLK
reset ~HRESETn

check_superlint -init
check_superlint -extract -top

file mkdir reports
check_superlint -report -violation -detailed -file reports/superlint_violations.txt -force
```

SPI Master Analysis

The SPI Master peripheral is an APB-interfaced IP. The analysis was run on August 30, 2026, using JasperGold 2025.06 FCS targeting the `apb_spi_master` top module.

Analyzed Source Files

The target design encompasses the following SystemVerilog files, loaded via the `jg_file.f` configuration:

- `apb_spi_master.sv`
- `spi_master_apb_if.sv`
- `spi_master_clkgen.sv`
- `spi_master_controller.sv`
- `spi_master_fifo.sv`
- `spi_master_rx.sv`
- `spi_master_tx.sv`

Violations Summary

The Superlint extraction yielded a total of **29 violations**. The design is free of hard errors. The breakdown is as follows:

- **CODINGSTYLE:** 18 total (15 Warnings, 3 Info)
- **STRUCTURAL:** 11 total (11 Warnings)

Important

Coding Style Observations: The majority of the coding style warnings (`ASG_MS_RTRU`, `ASG_NR_POVF`, `EXP_NR_MXSU`) stem from implicit 32-bit sizing on integer literals used in counter increments (e.g., adding `1` instead of `5'd1`). Additionally, a missing default state assignment (`CAS_NR_DEFX`) was flagged in the APB interface.

Raw Superlint Output (`superlint_violations.txt`)

The following listing contains the exact, unedited output from the JasperGold Superlint report generated for the SPI Master.

Listing 2: Raw Export of `superlint_violations.txt`

```
**************************************************
  Title: Jasper(TM) - Superlint Violation Report
**************************************************

  Date: Aug 30 10:51:22 IST 2026
  Tool Version: 2025.06 FCS 64 bits
  Top Module: apb_spi_master


********************************************************************************************
      Section 1: LINT DOMAIN
********************************************************************************************

============================================================================================
      Section 1.1: VIOLATIONS SUMMARY
============================================================================================
--------------------------------------------------------
 Category    | Error | Warning | Info | Waived | Total
--------------------------------------------------------
  CODINGSTYLE   0       15        3      0        18
  STRUCTURAL    0       11        0      0        11
  Total         0       26        3      0        29
```

```
============================================================================================
      Section 1.2: VIOLATIONS DETAILS  -  CODINGSTYLE
============================================================================================
--------------------------------------------------------------------------------------------------------------------

 Num  | Severity | Tag         | Message

                                                                                   | Source Location
                          | Instance-Module  | Grouping Status
--------------------------------------------------------------------------------------------------------------------

[1]     Warning    ASG_MS_RTRU   "Unequal length operands in assignment in design-unit/block '
    apb_spi_master'. Length of RHS is greater than LHS: LHS 'r_counter_tx' (unsigned) - 5 bit(s), RHS
    '(r_counter_tx + 1)' (unsigned) - 32 bit(s). 27 most significant bit(s) will be lost"   ../
    spi_master/apb_spi_master.sv :149      *-apb_spi_master   -
[2]     Warning    ASG_MS_RTRU   "Unequal length operands in assignment in design-unit/block '
    apb_spi_master'. Length of RHS is greater than LHS: LHS 'r_counter_rx' (unsigned) - 5 bit(s), RHS
    '(r_counter_rx + 1)' (unsigned) - 32 bit(s). 27 most significant bit(s) will be lost"   ../
    spi_master/apb_spi_master.sv :156      *-apb_spi_master   -
[3]     Warning    ASG_NR_POVF   "Potential overflow in assignment in design-unit/block '
    apb_spi_master'. LHS 'r_counter_tx' (unsigned) is of 5 bit(s), RHS '(r_counter_tx + 1)' (unsigned)
     can be of 6 bit(s) "                                                                          ../
    spi_master/apb_spi_master.sv :149      *-apb_spi_master   -
[4]     Warning    ASG_NR_POVF   "Potential overflow in assignment in design-unit/block '
    apb_spi_master'. LHS 'r_counter_rx' (unsigned) is of 5 bit(s), RHS '(r_counter_rx + 1)' (unsigned)
     can be of 6 bit(s) "                                                                          ../
    spi_master/apb_spi_master.sv :156      *-apb_spi_master   -
[5]     Warning    CAS_NR_DEFX   "Signal 's_int_tx' assigned in case statement is not assigned to X in
     default case"
                                                                                   ../spi_master/
    apb_spi_master.sv :172      *-apb_spi_master   Grouped
[6]     Warning    EXP_NR_MXSU   "Expression in design-unit 'apb_spi_master' has both signed '1' and
    unsigned 's_cnt_tx' expressions"
                                                                                    ../spi_master/
    apb_spi_master.sv :146      *-apb_spi_master   -
[7]     Warning    EXP_NR_MXSU   "Expression in design-unit 'apb_spi_master' has both signed '1' and
    unsigned 'r_counter_tx' expressions"
                                                                                    ../spi_master/
    apb_spi_master.sv :149      *-apb_spi_master   -
[8]     Warning    EXP_NR_MXSU   "Expression in design-unit 'apb_spi_master' has both signed '1' and
    unsigned 's_cnt_rx' expressions"
                                                                                    ../spi_master/
    apb_spi_master.sv :153      *-apb_spi_master   -
[9]     Warning    EXP_NR_MXSU   "Expression in design-unit 'apb_spi_master' has both signed '1' and
    unsigned 'r_counter_rx' expressions"
                                                                                    ../spi_master/
    apb_spi_master.sv :156      *-apb_spi_master   -
[10]    Warning    EXP_NR_MXSU   "Expression in design-unit 'apb_spi_master' has both signed '1' and
    unsigned 's_cnt_tx' expressions"
                                                                                    ../spi_master/
    apb_spi_master.sv :189      *-apb_spi_master   -
[11]    Warning    EXP_NR_MXSU   "Expression in design-unit 'apb_spi_master' has both signed '1' and
    unsigned 's_cnt_rx' expressions"
                                                                                    ../spi_master/
    apb_spi_master.sv :230      *-apb_spi_master   -
[12]    Warning    OPR_NR_UCMP   "Unequal length operands in equality operator encountered in design-
    unit/block apb_spi_master. LHS operand is 5 bits, RHS operand is 32 bits"
                                                                                    ../spi_master/
    apb_spi_master.sv :146      *-apb_spi_master   -
[13]    Warning    OPR_NR_UCMP   "Unequal length operands in equality operator encountered in design-
    unit/block apb_spi_master. LHS operand is 5 bits, RHS operand is 32 bits"
                                                                                    ../spi_master/
    apb_spi_master.sv :153      *-apb_spi_master   -
[14]    Warning    OPR_NR_UCMP   "Unequal length operands in equality operator encountered in design-
    unit/block apb_spi_master. LHS operand is 5 bits, RHS operand is 32 bits"
                                                                                    ../spi_master/
    apb_spi_master.sv :189      *-apb_spi_master   -
[15]    Warning    OPR_NR_UCMP   "Unequal length operands in equality operator encountered in design-
    unit/block apb_spi_master. LHS operand is 5 bits, RHS operand is 32 bits"
                                                                                    ../spi_master/
    apb_spi_master.sv :230      *-apb_spi_master   -
[16]    Info       MAC_NR_DMUL   "Macro 'log2' is defined with same definition"

                                                                                   ../spi_master/
    apb_spi_master.sv :11       *-apb_spi_master   -
[17]    Info       MAC_NR_DMUL   "Macro 'log2' is defined with same definition"

                                                                                   ../spi_master/
    spi_master_apb_if.sv :11    *-apb_spi_master   -
[18]    Info       MAC_NR_DMUL   "Macro 'log2' is defined with same definition"

                                                                                   ../spi_master/
    spi_master_fifo.sv :11      *-apb_spi_master   -


============================================================================================
      Section 1.3: VIOLATIONS DETAILS  -  STRUCTURAL
============================================================================================
--------------------------------------------------------------------------------------------------------------------
```

```
 Num  | Severity | Tag         | Message
                                | Source Location                      | Instance-Module
      | Grouping Status
---------------------------------------------------------------------------------------------------------------------

[1]      Warning    CMB_NR_TLIO   "Combinational path detected between top-level input 'PADDR[5:2]' and
     top-level output 'PRDATA'"    ../spi_master/apb_spi_master.sv:26    apb_spi_master-apb_spi_master
       -
[2]      Warning    CMB_NR_TLIO   "Combinational path detected between top-level input 'PADDR[5:2]' and
     top-level output 'PREADY'"    ../spi_master/apb_spi_master.sv:27    apb_spi_master-apb_spi_master
       Grouped
[3]      Warning    MOD_NO_IPRG   "Input port 'PADDR[5:2]' of top-level design-unit is not registered"
                                  ../spi_master/apb_spi_master.sv:21    apb_spi_master-apb_spi_master
       -
[4]      Warning    MOD_NO_IPRG   "Input port 'PENABLE' of top-level design-unit is not registered"
                                  ../spi_master/apb_spi_master.sv:25    apb_spi_master-apb_spi_master
       -
[5]      Warning    MOD_NO_IPRG   "Input port 'PSEL' of top-level design-unit is not registered"
                                  ../spi_master/apb_spi_master.sv:24    apb_spi_master-apb_spi_master
       -
[6]      Warning    MOD_NO_IPRG   "Input port 'PWDATA' of top-level design-unit is not registered"
                                  ../spi_master/apb_spi_master.sv:22    apb_spi_master-apb_spi_master
       -
[7]      Warning    MOD_NO_IPRG   "Input port 'PWRITE' of top-level design-unit is not registered"
                                  ../spi_master/apb_spi_master.sv:23    apb_spi_master-apb_spi_master
       -
[8]      Warning    MOD_NO_IPRG   "Input port 'spi_sdi0' of top-level design-unit is not registered"
                                  ../spi_master/apb_spi_master.sv:42    apb_spi_master-apb_spi_master
       -
[9]      Warning    MOD_NO_IPRG   "Input port 'spi_sdi1' of top-level design-unit is not registered"
                                  ../spi_master/apb_spi_master.sv:43    apb_spi_master-apb_spi_master
       -
[10]     Warning    MOD_NO_IPRG   "Input port 'spi_sdi2' of top-level design-unit is not registered"
                                  ../spi_master/apb_spi_master.sv:44    apb_spi_master-apb_spi_master
       -
[11]     Warning    MOD_NO_IPRG   "Input port 'spi_sdi3' of top-level design-unit is not registered"
                                  ../spi_master/apb_spi_master.sv:45    apb_spi_master-apb_spi_master
       -


********************************************************************************************
```

Engineering Analysis & Waiver Justifications

An exhaustive review of the 29 flagged violations confirms that they are artifacts of generalized linting rules conflicting with standard, intentional hardware practices. All listed violations are safe to waive.

Counter Width Truncations (ASG_MS_RTRU, ASG_NR_POVF)

The tool flags `r_counter_tx` and `r_counter_rx` because the un-sized integer literal `1` is implicitly treated as a 32-bit signed integer. During the increment operation, the 5-bit counter is temporarily promoted to 32 bits and then truncated back down to 5 bits upon assignment. Because this is a standard wrap-around counter, discarding the overflow bit is the exact intended behavior and safely optimized by synthesis.

Signed/Unsigned Mix & Operand Mismatches (EXP_NR_MXSU, OPR_NR_UCMP)

Similar to the truncation warnings, these flags occur when verifying if the counter has reached its target (e.g., `r_counter_tx == s_cnt_tx - 1`). The subtraction of the literal `1` promotes the right-hand side expression to a 32-bit signed integer, causing a mismatch against the 5-bit unsigned register. This is a harmless artifact that the synthesizer resolves.

Macro Redefinition (MAC_NR_DMUL)

The `log2` macro is flagged for being defined multiple times. The exact same definition is included at the top of the top-level module, the APB interface, and the FIFO modules. Because the definitions are completely identical, they do not overwrite each other with conflicting logic and pose zero risk to the design.

Case Statement Default (CAS_NR_DEFX)

The linter warns that the `s_int_tx` signal is not explicitly assigned to an unknown state (`X`) inside the `default` branch of the state machine. However, a default safe assignment (`s_int_tx = 1'b0`) is made directly before the `case` block opens. This pre-assignment guarantees no latches will be inferred, making the hardware perfectly safe.

Structural Boundary Flags (CMB_NR_TLIO, MOD_NO_IPRG)

The structural engine flags combinational paths from the input address to the read data/ready outputs, as well as unregistered APB control signals and SPI data lines. In the APB protocol, the slave utilizes combinational address decoding to drive `PRDATA` and assert `PREADY` immediately for zero-wait-state responses. Registering these specific inputs at the boundary would violate the APB handshake timing. These are intentional architectural choices for a standard bus peripheral.